

K2 Phase 1 (Lessons 1–10)

Lesson	Title	Lesson Objectives	Tool(s) on Platform	Additional Details
1	"AI, Meet My Shapes!"	Teach AI to recognize 3 shapes (circle, square, triangle) via drawing. Test if AI remembers.	Drawing recognition tool, voice test ("Do you know what shape this is?")	Students draw each shape 3x. AI starts with answering "No, I don't know what shape this is" and ends saying "Yes, that's a circle."
2	"Robot Builder"	Review shapes with AI. Generate a simple robot using shape instructions (e.g., "Draw a square body, circle head").	Drawing recognition + voice-to-text and Voice-to-text and text-to-image generation	Teacher models instruction; kids give 2-step commands.
3	"Color the Rainbow"	Teach AI 3 primary colors (red, blue, yellow) by coloring shapes on tablet.	Drawing + coloring recognition tool	Students can play a color recognition game to review.
4	"Color My Robot"	Review colors. Generate a colored robot with instruction ("Make my robot's head red and its body blue").	Voice-to-text and text-to-image generation + color review quiz	
5	"Nature Hunt"	Teach AI nature items (sun, cloud, flower) by pointing to real photo (e.g., "This is a flower").	Image part-pointing recognition tool	
6	"Animals Stamp & Test"	Review nature Teach AI animals with toys & picture cards Quick Game: AI tests students: AI says "Show me a duck," student finds correct play-doh stamp.	Image recognition + voice quiz	

7	"Invent a Silly Animal"	Review Game: AI tests students: AI says "Show me a duck," student finds correct play-doh stamp. Students invent hybrid (e.g., "duck with cat ears") via drawing/play-doh → teach AI → AI generates it.	Image recognition + text-to-image	
8	"AI's Feeling Cards"	Teach AI emotions (happy, sad, angry) using picture cards with faces. Play matching game: AI draws the face, students guess.	Image recognition (cards), emotion game mode	AI learns from 3 card examples per emotion.
9	"Add a Smile"	Review emotions. Teach AI emotions via drawing faces. Generate a face on the robot ("Make robot happy").	Drawing recognition + Voice-to-text and text-to-image	Kids draw 3 different happy faces; AI generalizes.
10	"Let's Throw AI a Birthday Party!"	Kids tell AI what to put in the birthday picture: Balloons (circle shapes) Presents (square + triangle shapes) Birthday robot (color it red, blue, yellow) Outside: sun, cloud, flower A silly animal guest (invent one) The robot's face: happy or silly AI generates the whole party picture.	Voice-to-text and text-to-image Can review a few previous tools to test AI before the main activity	

K2 Phase 2 (Lessons 11–20)

Lesson	Title	Lesson Objectives	Tool(s) on Platform	Additional Details
11	“Apple, Banana, Bear”	Teach AI fruits (apple, banana) and one animal (bear) using real toys. Test by showing & asking.	Image recognition (toy camera mode)	
12	“Play-Doh Fruits”	Review fruits. Students make a fruit with play-doh, show AI, AI names it.	Image recognition	Cross-medium learning (3D → digital recognition).
13	“Jump, AI, Jump!”	Teach AI body movements: stand, jump, clap (joint recognition). Kids demonstrate, AI repeats name.	Joint recognition tool	AI says “That is a jump” after 3 examples.
14	“Dance with Me”	Review movements. Teach AI a simple 2-step dance (clap → jump). AI performs back via avatar.	Joint recognition + playback	
15	“What’s That Sound?”	Teach AI sounds (clap, stomp, “wow”). Record, label, AI repeats sound back.	Sound recording & playback tool	
16	“Make a Beat”	Students record 3 sounds, press buttons to arrange into a simple 4-beat song. AI generates song from arrangement.	Songmaker tool (button-based)	
17	“Point & Learn”	Teach AI body parts (head, shoulders, knees, toes) via pointing on a body diagram.	Image part-pointing (body template)	
18	“AI’s Turn to Lead”	AI gives movement commands (e.g., “touch knees”), students follow.	Joint recognition + voice instruction	Simple commands only.

19	"Charades with AI"	Teach AI animal actions through joint recognition. Play animal charades. Student acts (e.g., frog jump). AI guesses using joint+movement recognition.	Joint recognition + animal library	
20	"Animal Dance Party"	Choose an animal from Phase 1. Generate that animal leading a simple dance (AI animates the animal).	Joint recognition + animal generation	No avatar – student-chosen animal.

K3 Phase 1 (Lessons 1–20)

Lesson	Title	Lesson Objectives	Tool(s) on Platform	Additional Details
1	“Zoo Visit”	Teach AI 3 new animals using toys + play-doh stamps. (e.g., giraffe, zebra, elephant)	Image recognition (toys + moulds)	
2	“Follow the Pattern”	Play pattern recognition game (shapes/colors) to review. AI tests students: “What comes next?”	Pattern game mode	
3	“AI Architect”	Generate a house using shapes + 2 colors (e.g., “star window, red roof”).	Voice-to-text and text-to-image generation	
4	“Emotion Mix-Up”	Teach AI new emotions: scared, surprised. Use drawing + picture card.	Drawing + image recognition	
5	“Arrange the Feelings”	Arrange 3 picture scenes (e.g., lost toy → sad → find toy → happy). AI generates 1 sentence per scene.	Story scene arrangement + text generation	
6	“Left, Right, Up, Down”	Teach AI finger directions. Combine into a pattern (left → up → right).	Finger recognition	
7	“My Own Instrument”	Record 5 sounds (voice, clap, tap table). Arrange into 8-beat song. AI generates melody.	Songmaker	
8	“Fingers-and-Joints Dance”	Teach AI finger directions (left, right, up, down) plus joint movements. Combine into 4-step dance.	Finger + joint recognition	E.g., “point left → jump → point right → clap.”

9	"Tell Me a Story"	Generate a 3-sentence story: character (robot), setting (garden), feeling (happy).	Text generation + image generation (one scene)	
10	"Instant Story Time"	Students generate one scene; AI describes it in 1 sentence.	Drawing recognition + Voice-to-text and text-to-image	

K3 Phase 2 (Lessons 11–20)

Lesson	Title	Lesson Objectives	Tool(s) on Platform	Additional Details
11	“Numbers on Fingers”	Teach AI numbers 6–10 using fingers. Play counting game: AI flashes fingers, kids say number.	Finger recognition + number quiz	
12	“Play-Doh Fruits”	Students make a fruit with play-doh, show AI, AI names it.	Image recognition	Cross-medium learning (3D → digital).
13	“Fruit Animal”	Generate a hybrid: “Make a bear with an apple body.”	Text-to-image + drawing recognition	
14	“Arrange the Story”	Arrange 4 story scenes. AI generates 4 sentences total.	Story scene + text generation	Class votes on best AI story.
15	“Make a Monster”	Generate a monster using a few descriptive words (color, texture, body parts, emotion).	Voice-to-text and text-to-image generation	
16	“Step, Step, Stop”	Teach AI a sequence using joint recognition (step left, step right, stop, wave).	Joint recognition + sequence recording	
17	“AI Music Video”	Combine song + dance. AI generates a full 30-second performance.	Songmaker + joint playback + Voice-to-text and text-to-image (background)	
18	“What Am I?”	Play with AI: describe an object (e.g., sun, duck, apple). AI must guess within 3 tries.	Drawing recognition + timer mode	

19	"Happy Feet, Angry Stomp"	Teach AI to connect emotion → movement (happy = bouncy steps, angry = stomp).	Joint recognition + emotion library	Kids select emotion, AI performs movement.
20	"AI & Me: The Movie"	Generate after full scene: character, setting, feeling, actions, 2 objects, 1 sentence story. Then AI animates it.	Full platform integration	